

Air University



Organization & Assembly (Lab)

Year 2025

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Academic Year: 2024 – 2028

Department of Cyber Security

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BS – Cyber Security – Fall – 24 (A)

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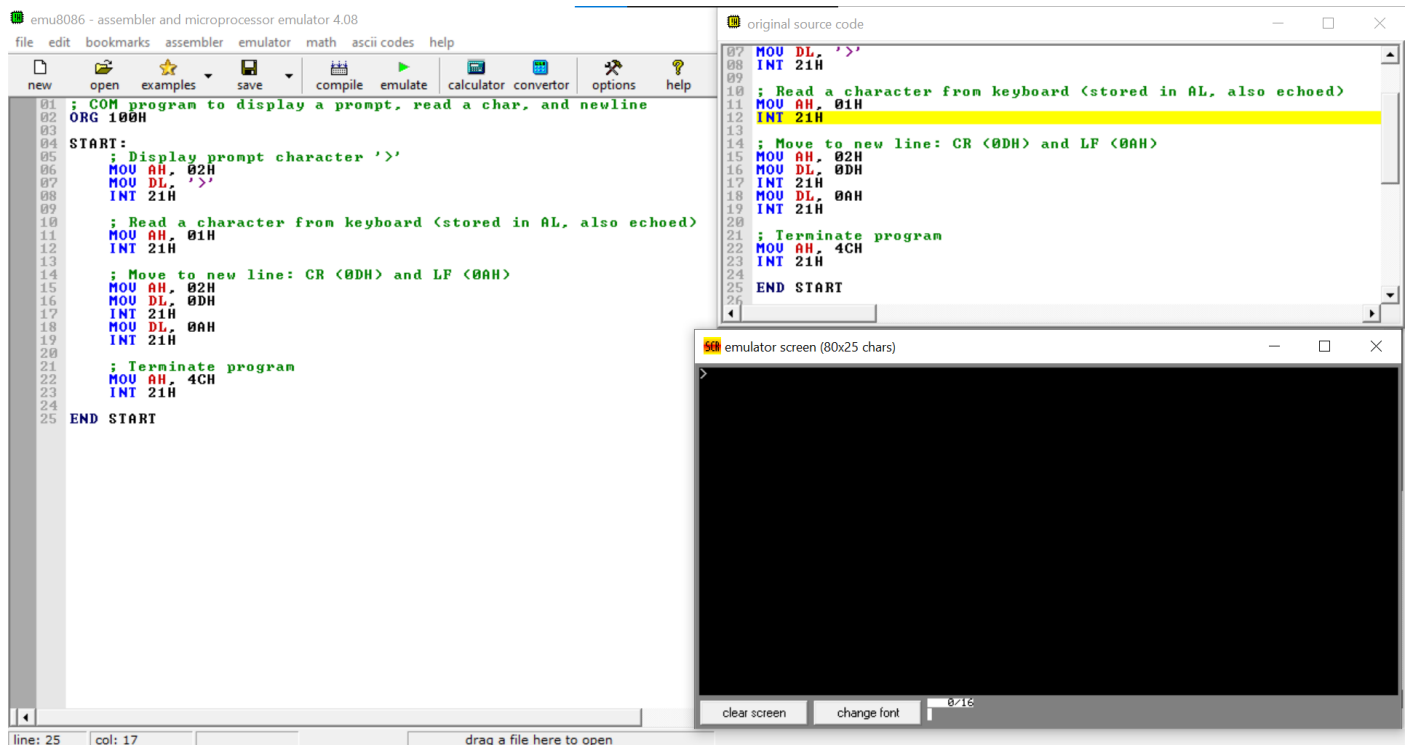
Lab Task # 06

Write a program in assembly language using COM and EXE templates to display a character, getting a character from a user, and moving to a new line.

COM Template:

Code & Output

- The program prompted for input, which can be seen in the black screen.



```
01 ; COM program to display a prompt, read a char, and newline
02 ORG 100H
03
04 START:
05 ; Display prompt character '>'
06 MOV AH, 02H
07 MOV DL, '>'
08 INT 21H
09
10 ; Read a character from keyboard (stored in AL, also echoed)
11 MOV AH, 01H
12 INT 21H
13
14 ; Move to new line: CR (0DH) and LF (0AH)
15 MOV AH, 02H
16 MOV DL, 0DH
17 INT 21H
18 MOV DL, 0AH
19 INT 21H
20
21 ; Terminate program
22 MOV AH, 4CH
23 INT 21H
24
25 END START
```

```
07 MOV DL, '>'
08 INT 21H
09
10 ; Read a character from keyboard (stored in AL, also echoed)
11 MOV AH, 01H
12 INT 21H
13
14 ; Move to new line: CR (0DH) and LF (0AH)
15 MOV AH, 02H
16 MOV DL, 0DH
17 INT 21H
18 MOV DL, 0AH
19 INT 21H
20
21 ; Terminate program
22 MOV AH, 4CH
23 INT 21H
24
25 END START
```

emu8086 - assembler and microprocessor emulator 4.08

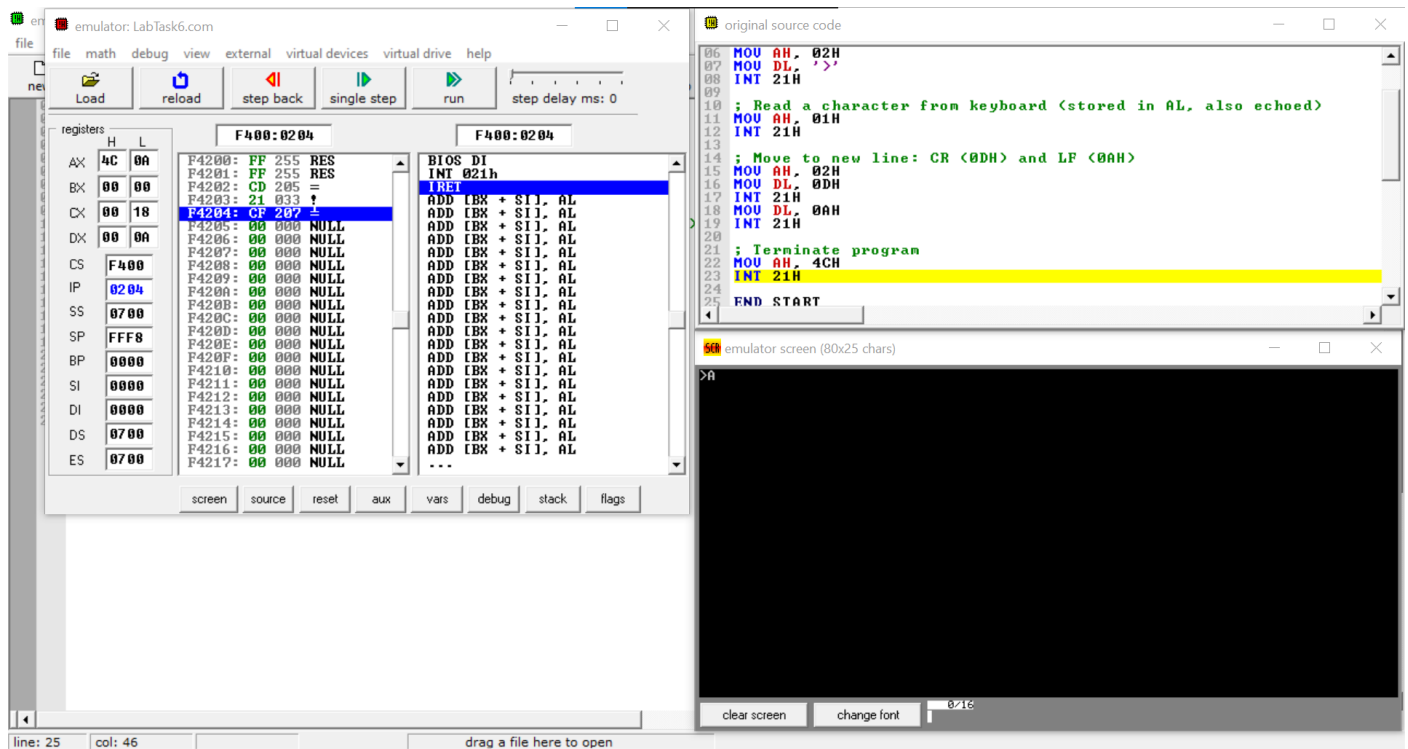
original source code

emulator screen (80x25 chars)

clear screen change font 0/16

line: 25 col: 17 drag a file here to open

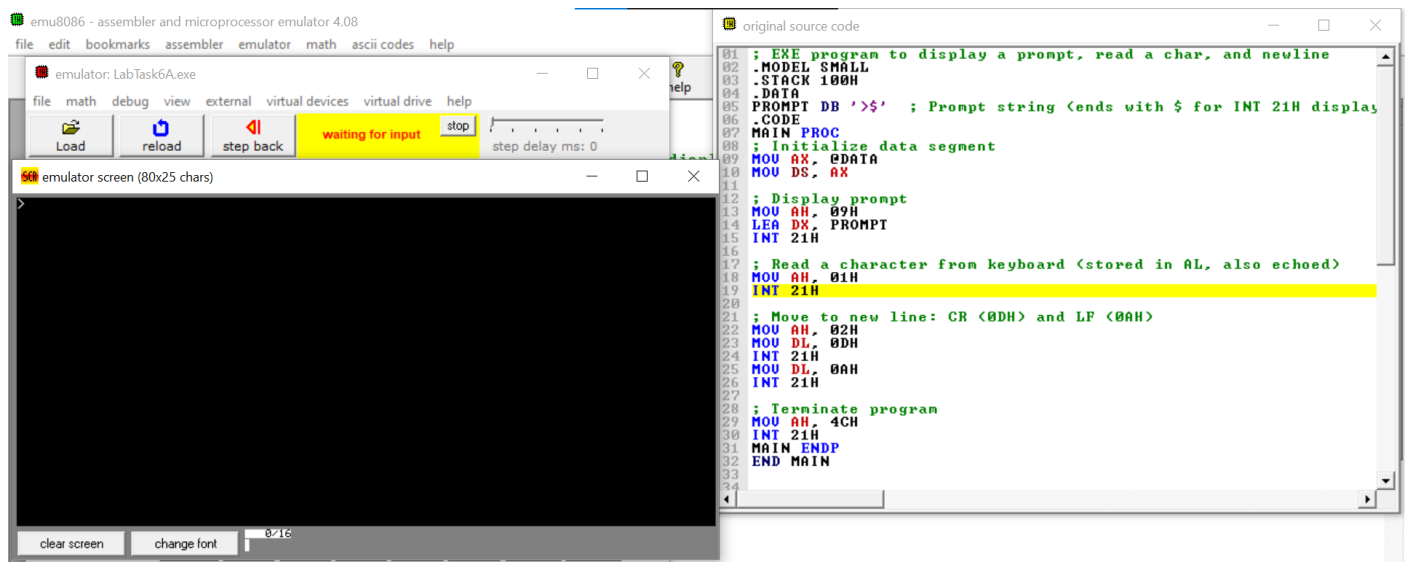
- The program accepted the input and updated the **AX** register with input value.



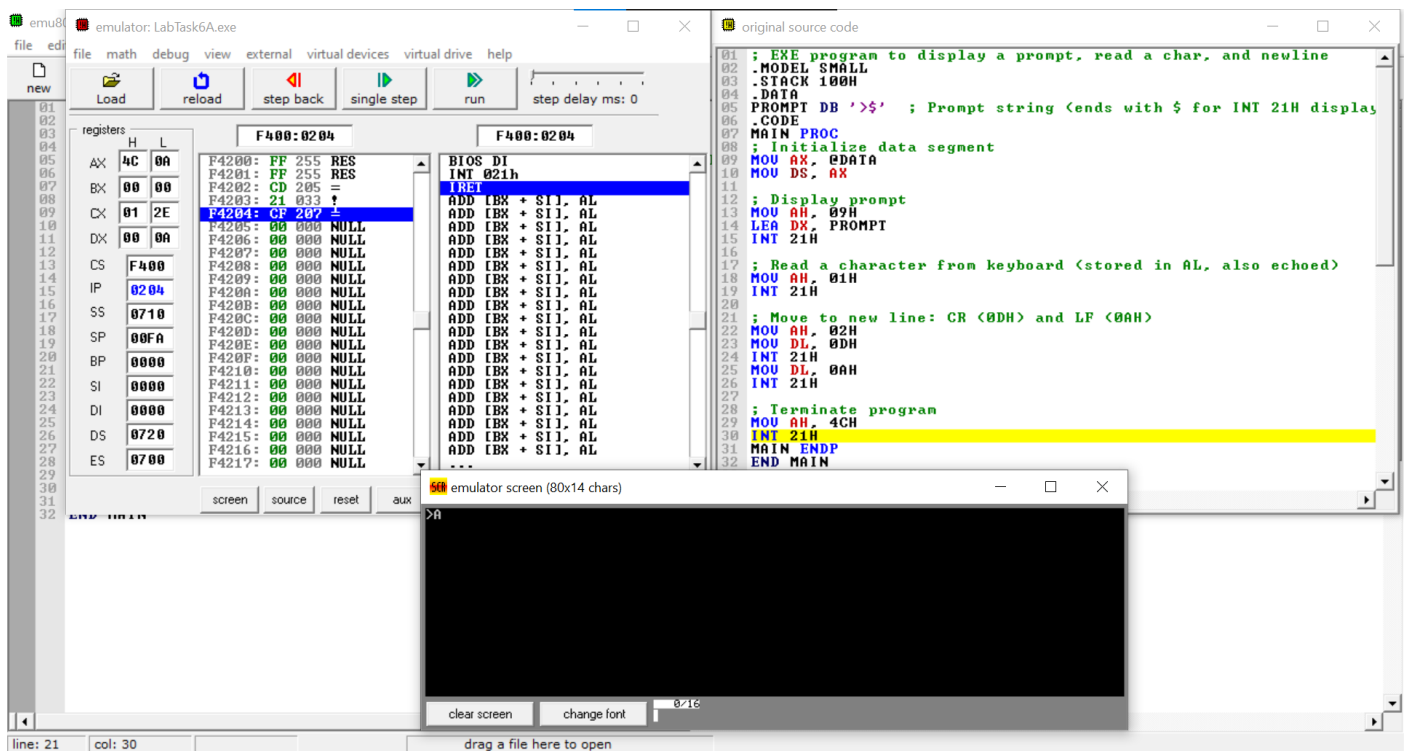
EXE Template:

Code & Output

- The program prompted for input, which can be seen in the black screen.



- The program accepted the input and updated the **AX** register with input value.



THE END